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Measuring Multidimensional Resilience of China's Oil and Gas Industry and Forecasting Resilience Under Multiple Scenarios

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Abstract

In the context of a rapidly changing global energy landscape and mounting pressures on energy security, enhancing the resilience of the oil and gas industry (OGI) has become a critical task for safeguarding China's energy security. This study develops a multidimensional resilience indicator system—comprising recovery, adaptability, responsiveness, and innovation—and, based on OGI data for 2001–2022, employs the entropy method to quantitatively assess resilience by sub-dimension and development stage. Leveraging a backpropagation (BP) neural network, we construct a dynamic simulation model to produce long-term, multi-scenario forecasts of China's OGI resilience for 2023–2032, enabling comparison of development potential across scenarios. The results indicate that overall resilience exhibited a fluctuating upward trend and reached a medium-strength resilience level by 2022, with innovation and recovery gradually emerging as the dominant drivers. Forecasts show that under the green-transition scenario, resilience will improve the most, increasing by 5.49% by 2032 and reaching the threshold for strong resilience earlier than under other scenarios. These findings offer actionable insights for enhancing the reliability and sustainability of energy supply chains in the face of climatic and geopolitical challenges.

Keywords: oil and gas industry; resilience; scenario simulation; BP neural network; sustainable development

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1. Introduction

As the core component of the national energy system, the oil and gas industry (OGI) serves as a critical domain for ensuring national energy security and as a cornerstone for stabilizing the national economy; accordingly, its resilience is directly linked to both energy security and economic stability. The report of the 20th National Congress of the Communist Party of China underscores the imperative of enhancing the resilience and security of industrial and supply chains. The Third Plenary Session of the 20th Central Committee further emphasizes the necessity of establishing robust institutional mechanisms to uplift the resilience and security of industrial and supply chains and promotes building an autonomous and controllable OGI chain. In recent years, a confluence of factors—including geopolitical conflicts, extreme climate events, technological transformation, and energy transition—has heightened the uncertainties confronting the domestic OGI. Simultaneously, the domestic OGI's dependence on foreign supplies has continued to rise, with crude oil import reliance exceeding 72% and natural gas import reliance approaching 45% [1]. Against this backdrop, evaluating the resilience of the OGI and conducting scenario-based forecasting have emerged as central concerns in both academic and industrial circles. This

study, grounded in the practical imperatives of the OGI and resilience research, aims to employ quantitative analyses and scenario simulations to explore the industry's risk-response capabilities under different scenarios, thereby offering insights for the sustainable development of China's OGI in the future.

The word resilience derives from the Latin verb resilio, which denotes the ability to return an object to its original state [2]. Following the Industrial Revolution in the late 19th century, the concept of resilience was first introduced into mechanics and materials science, where it referred to the capacity of materials to adapt to and recover from external forces. By the mid-20th century, psychology adopted the term to describe an individual's capacity to adapt under stress. C. S. Holling (1973) extended the concept to ecological systems, defining resilience as "the capacity of a complex system to adapt to external shocks, absorb disturbances, and continue to recover, maintaining inherent stability and persistence" [3]. With the advancement of ecological resilience studies, scholars gradually proposed the notion of evolutionary resilience, which emphasizes that resilience is not merely the ability to return to an initial state, but also the capacity of a system to change, adapt, and transform under pressures and constraints, embodying persistence, adaptability, and transformability [4]. Pimm (1984) further argued that resilience is a multidimensional process encompassing system stability, recovery capacity, and transformational flexibility [5]. Within the socio-ecological system framework, Walker et al. (2004) highlighted that resilience extends beyond resistance to stress; rather, it is a dynamic process comprising three phases—recovery, adaptation, and transformation [6]. This theoretical evolution, shifting from static recovery to a dynamic concept stressing system flexibility, has laid the foundation for subsequent research across multiple disciplines.

In the energy domain, resilience generally refers to the capacity of energy systems to respond to emergencies and, in particular, to maintain stable operation under high-risk and complex market conditions. Scholars have noted that resilience in the energy sector can be continuously enhanced through measures such as technological innovation and policy adaptation. Hromada et al. (2021) proposed a Critical Resilience Assessment (CRA) method for electricity critical infrastructure that integrates physical security, cybersecurity, and other factors, enabling comprehensive monitoring and evaluation of the resilience of key grid components [7]. Li Kun and Cao Xiabing (2021) examined the formation mechanisms of digital resilience in energy firms and emphasized the role of digital transformation in strengthening corporate resilience. Research on energy-sector resilience focuses on improving systems' fault tolerance, adaptive capacity, and recovery speed in the face of internal and external shocks, which is crucial for safeguarding energy supply security and advancing the energy transition [8]. Within the OGI, Scholars have explored resilience assessment and enhancement pathways from multiple dimensions. Mahmood (2023) systematically articulated an integrated approach to sustainability, reliability, and resilience (the "3R" framework) in the design and construction of oil and gas infrastructure [9]. Hosseini et al. (2016) developed a network-theory-based framework for assessing OGI resilience by quantifying vulnerabilities across supply-chain stages and thereby estimating the industry's capacity to withstand disruptions such as supply-chain interruptions [10]. More recently, Sujan Piya (2024) employed a fuzzy AHP–TOPSIS methodology to construct a green supply-chain indicator system, identifying risk management capability and the strength of government regulation as key drivers of resilience in oil and gas supply chains [11].

As research has progressed, methods for constructing resilience indicator systems, intelligent modeling, and scenario forecasting have been strengthened. Existing studies have quantified resilience by developing composite indicator frameworks and have employed machine learning (ML) and artificial intelligence (AI) techniques to improve

predictive accuracy. Duan et al. (2023) used an Exponential Generalized Autoregressive Conditional Heteroskedasticity (EGARCH) model to quantify and analyze risks in China's carbon market, offering new approaches for quantitative risk assessment [12]. Zhao Lizhou and Zhang Ningfeng (2024) developed a petrochemical-enterprise supply-chain resilience assessment model based on the analytical hierarchy process and a BP neural network to evaluate firms' supply-chain resilience under the dual-circulation strategy [13]. Balan et al. (2025) reviewed AI and ML applications in supply-chain resilience, noting that ML methods can process large volumes of data to provide real-time insights into supply-chain states, reduce demand-forecast errors by roughly 10–20%, and accelerate disruption response by approximately 20–30%, thereby enhancing resilience [14]. A systematic review by Ayadi et al. (2025) indicates that ML algorithms can be used to predict extreme weather events such as floods, heatwaves, and hurricanes, providing early warning and decision support for resilience enhancement [15]. Chen et al. (2023) embedded a BP neural network within a multi-scenario ecological extended input–output model, demonstrating a feasible pathway for scenario-driven, system-level quantitative forecasting using BP neural networks [16]. Collectively, the literature increasingly emphasizes integrating ML techniques with resilience indices to construct more accurate resilience assessment systems based on big data and model-based forecasting, thereby supporting sectoral decision-making.

In summary, following the conceptual evolution through engineering resilience, ecological resilience, and evolutionary resilience, the concept of resilience has gradually been incorporated into the field of industrial economics [17]. Existing studies have provided a robust foundation for both theoretical refinement and methodological development of industrial resilience. However, the theoretical framework for industrial resilience remains underdeveloped, and quantitative evaluation methods are still in their nascent stages. Under the dual pressures of energy security and economic development, it is increasingly important to enhance the resilience of the OGI—improving its capacity to respond to and recover from external shocks. Building on existing theory, the present study reviews extant resilience constructs and, accounting for the characteristics of the OGI, establishes a multidimensional resilience framework encompassing recovery, adaptability, responsiveness, and innovation capacity. We employ entropy-weighted metrics in conjunction with a BP neural network to quantitatively assess and dynamically forecast industrial resilience. Furthermore, we design multiple development scenarios to simulate future resilience trajectories. The study not only enriches the application of resilience theory at the industry level but also provides a practical path for resource-based industries to optimize safety governance and achieve high-quality sustainable development.

2. Research Methods

2.1. Entropy Method

The entropy method is an objective weighting method based on information entropy theory, which is often used to determine the weight of each indicator in multi-indicator comprehensive evaluation problems. It evaluates the degree of variation of each indicator: the larger an indicator's variation, the greater its importance, while information entropy is inversely related to variation. The higher the impact of an indicator on the resilience of the OGI, the greater its weight, and vice versa [18]. Using the entropy method to determine the weight of each indicator can not only reasonably avoid the subjective influence caused by artificial conjecture but also effectively overcome the problem of information overlap between multiple indicator variables [19]. This paper adopts the entropy method to calculate the weights of 24 evaluation indicators that affect the resilience of China's OGI.

To eliminate scale effects, the original data are first normalized. For positive indicators, the normalization formula is

$$x'_{ij} = \frac{x_{ij} - min\{x_{1j}, \cdots, x_{nj}\}}{max\{x_{1j}, \cdots, x_{nj}\} - min\{x_{1j}, \cdots, x_{nj}\}} \tag{1}$$

For the negative indicator, the normalization formula is

$$x'_{ij} = \frac{max\{x_{1j}, \cdots, x_{nj}\} - x_{ij}}{max\{x_{1j}, \cdots, x_{nj}\} - min\{x_{1j}, \cdots, x_{nj}\}} \tag{2}$$

The weight of each indicator in year is

$$p_{ij} = \frac{x'_{ij}}{\sum_{i=1}^{n} x'_{ij}} \tag{3}$$

When $x'_{ij} = 0$, $p_{ij}\ln(p_{ij}) = 0$ is set to avoid infinity.

The information entropy of each indicator is

$$e_j = -k\sum_{i=1}^{n} p_{ij}\ln(p_{ij}) \tag{4}$$

Information entropy. The larger the value, the more uniform the information distribution of the indicator and the weaker the distinguishing ability.

The Redundancy (also called the Coefficient of Variation) is

$$d_j = 1 - e_j \tag{5}$$

The indicator weight is

$$w_j = \frac{d_j}{\sum_{j=1}^{m} d_j} \tag{6}$$

After obtaining the objective weights of each indicator, the comprehensive resilience score of the sample is calculated according to the following formula:

$$S_i = \sum_{j=1}^{m} w_j \cdot x'_{ij} \tag{7}$$

Finally, the composite resilience score for China's OGI was obtained. This score reflects the temporal trend of overall resilience and the relative contribution of each subdimension, and provides a quantitative basis for subsequent BP neural network modeling and scenario forecasting.

2.2. BP Neural Network

A BP neural network (backpropagation neural network) is a multi-layer feedforward neural network trained based on the error backpropagation algorithm. Its basic structure consists of an input layer, multiple hidden layers, and an output layer. Each layer contains several neuron nodes [20]. The network architecture is shown in Figure 1.

In this paper, we adopt a BP neural network to predict China's OGI resilience. The newff function in MATLAB R2024b is used to construct a three-layer BP neural network. A total of 24 indicators in the recovery, adaptability, responsiveness, and innovation subsystems and the weights calculated by the entropy method are used as input layer data, and the OGI resilience score is used as output layer data. Based on the empirical formula, the number of hidden layer nodes $\mathrm{L} = \sqrt{(\mathrm{m} + \mathrm{n}) + \mathrm{a}}$ is determined to be 7 nodes, and a

BP neural network model with a topological structure of 24 × 7 × 1 is constructed [21]. The input layer neurons are X1~X24, the hidden layer neurons are 7, and the output layer neuron is 1, with stationarity determined using the ADF unit root test. The learning rate is set to 0.01, the error threshold is set to 1×10^{-6}, and the maximum number of training times is set to 1000 times. The random seed is fixed using rng(2222), and training is terminated once the error converges to the predefined threshold.

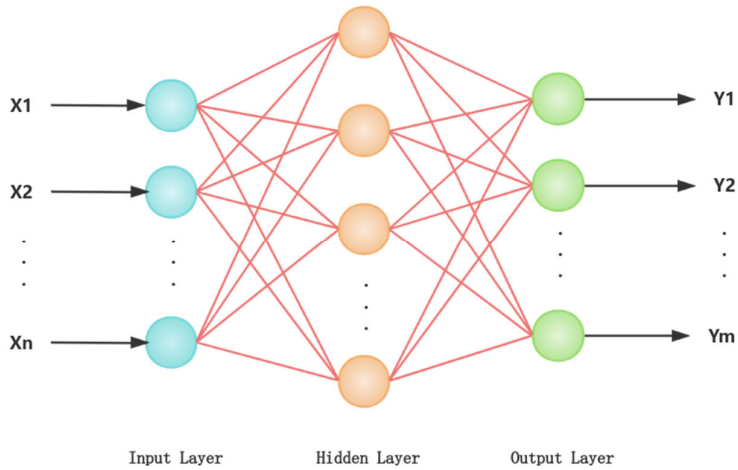

Figure 1. BP neural network structure.

Suppose a pair of samples (*X*,*Y*) is $X = [x_1, x_2 \ldots x_m]^T$, $Y = [y_1, y_2 \ldots y_n]^T$ and the hidden layer neurons are $O = [o_1, o_2, o_l]^T$; the network weight matrix W1 between the input layer and the hidden layer, and the network weight matrix W2 between the hidden layer and the output layer are, respectively, $w^1 = \begin{bmatrix} w^1_{11} & \cdots & w^1_{1m} \\ \vdots & \ddots & \vdots \\ w^1_{l1} & \cdots & w^1_{lm} \end{bmatrix}$, $w^2 = \begin{bmatrix} w^2_{11} & \cdots & w^2_{11} \\ \vdots & \ddots & \vdots \\ w^2_{n1} & \cdots & w^2_{n1} \end{bmatrix}$, and the thresholds of the hidden layer neurons and the thresholds of the output θ^1 layer neurons θ^2 are $\theta^1 = [\theta^1_1, \theta^1_2, \cdots \theta^1_l]'$, $\theta^2 = [\theta^2_1, \theta^2_2 \cdots \theta^2_n]'$.

The output of the hidden layer neurons is shown in Equation (8):

$$O_j = f\left(\sum_{i=1}^{m} w^1_{ji} x_i - \theta^1_j\right) \tag{8}$$

where $f(\cdot)$ is the transfer function of the hidden layer, which is used to change the output of the neuron. In this paper, the hyperbolic tangent *S*-shaped transfer function $f(x) = \frac{1}{1+e^{-2n}} - 1, (-1 < f(x) < 1)$ is used.

The output of the output layer neurons is shown in Equation (9):

$$z_k = g\left(\sum_{j=1}^{l} w^2_{kj} o_j - \theta^2_k\right) \tag{9}$$

where $g(\cdot)$ is the transfer function of the output layer, and the linear transfer function $g(x) = x$ is used in this paper.

3. Construction of Indicator System and Data Sources

3.1. Construction of Indicator System

Based on the full consideration of the characteristics of the OGI, this paper draws on relevant research results in other fields [22–25], uses the entropy method to calculate

weights, and constructs an OGI resilience index system that includes four dimensions: recovery, adaptability, responsiveness, and innovation, as shown in Table 1.

Table 1. Comprehensive evaluation index system and weights of China's OGI resilience.

Dimensions	Index	Property	Weight
Recovery (0.3125)	OGI Capital Stock	Positive Correlation	0.0493
	Gross Domestic Product	Positive Correlation	0.0625
	Number of OGI Enterprises	Positive Correlation	0.0638
	Oil (Gas) Pipeline Mileage	Positive Correlation	0.0466
	Pipeline Freight Volume	Positive Correlation	0.0473
	Energy Industry Investment	Positive Correlation	0.0430
Adaptability (0.1929)	Oil's Share of Total Energy Consumption	Negative Correlation	0.0252
	Estimated Ultimate Recovery of Oil	Positive Correlation	0.0468
	Oil Import Dependence	Negative Correlation	0.0424
	Total Asset Profit Rate of Industrial Enterprises Above Designated Size	Positive Correlation	0.0251
	Efficiency of Energy Conversion	Positive Correlation	0.0249
	Oil Production	Positive Correlation	0.0285
Responsiveness (0.1602)	Elasticity Ratio of Energy Production	Positive Correlation	0.0567
	International Oil Price Fluctuations	Negative Correlation	0.0169
	Oil Supply and Demand Gap	Negative Correlation	0.0140
	Carbon Emissions	Negative Correlation	0.0138
	Non-fossil Energy Consumption Share	Negative Correlation	0.0286
	Per Capita Energy Consumption	Positive Correlation	0.0302
Innovation (0.3344)	CCUS Technology Adds New CO_2 Processing Capacity	Positive Correlation	0.1119
	Full-time Equivalent Input of R&D Personnel in the OGI	Positive Correlation	0.0306
	Number of Invention Applications and Patent Applications in the OGI	Positive Correlation	0.0753
	Funding for R&D Investment OGI to Develop New Products	Positive Correlation	0.0440
	Investment in Industrial Pollution Control Completed	Positive Correlation	0.0424
	Number of Graduates with College Degree or Above	Positive Correlation	0.0302

Recovery resilience denotes the ability to withstand risks and recover rapidly. Drawing on Zhang Wei et al. [26], this study selects indicators across two dimensions—financial performance and investment—focusing on three key metrics: OGI capital stock, gross domestic product (GDP), and energy-sector investment. OGI capital stock measures the industry's asset base, while GDP reflects macroeconomic conditions supporting recovery. Additionally, logistics-support indicators—such as oil and gas pipeline length and pipeline freight volume—are included to capture transportation continuity during disruptions [27,28].

Adaptability resilience [22] denotes the capacity of the OGI to maintain core functions and pursue sustainable development under shocks by dynamically reallocating resources, optimizing technological trajectories, and improving economic efficiency. We assess this dimension using six indicators: share of oil in total energy consumption, estimated ultimate recovery of oil, oil import dependence, return on total assets of industrial enterprises above designated size [29], energy conversion efficiency, and oil production. These indicators measure the industry's ability to preserve functional stability and adapt.

Responsiveness resilience encompasses the system's agility to flexibly adapt to environmental changes, rapidly address risks, and mitigate damage, as well as its collaborative capacity. This study utilizes three primary indicators—elasticity ratio of energy production, international oil price fluctuations, and oil supply and demand gap—to reflect the system's ability to adjust output, its sensitivity to external market disturbances, and the degree of supply–demand imbalance [30]. Furthermore, we incorporate carbon emissions, non-fossil energy consumption share, and per capita energy consumption to capture the OGI's response efficiency in environmental adaptation and energy-structure transformation [31]. Together, these multidimensional metrics enable a comprehensive assessment of its ability to react effectively in risk management and dynamic resource allocation.

Innovation resilience refers to the capacity of the OGI to sustain and enhance its competitiveness in the face of internal and external challenges through technological innovation, talent reserves, and capital investment. Scientific and technical personnel serve as the intrinsic driving force behind the industry's sustainable development. To assess the strength of talent reserves, this study adopts two evaluation indicators: the full-time equivalent input of research and development (R&D) personnel in the OGI and the number of graduates with a college degree or above [32]. At the same time, the number of invention and patent applications in the OGI, as well as funding for R&D investment to develop new products, are used to represent technological innovation and financial investment capabilities. In addition, CCUS, as a "game-changer" technology that not only reduces carbon emissions but also fosters technological upgrading [33,34], green innovation, and resilience of resource-dependent industries, was assigned the largest weight. CCUS capacity is consistent with its strategic role in aligning the OGI with low-carbon transition pathways.

3.2. Data Source and Processing

The primary indicator data for 2001–2022 were obtained from the China Statistical Yearbook, China Energy Statistical Yearbook, and the annual OGI Development Report. Missing values were imputed using linear interpolation.

4. Results Analysis

4.1. Resilience Evaluation

Combined with the research on the classification of resilience [35], the Natural Breaks is used to divide the resilience (R) of China's OGI into five levels: level I low resilience ($0 < R < 0.3$), level II medium-low resilience ($0.3 \leq R < 0.45$), level III medium resilience ($0.45 \leq R < 0.7$), level IV medium-strength resilience ($0.7 \leq R < 0.85$), and level V strength resilience ($0.85 \leq R \leq 1.0$), as shown in Table 2.

Table 2. Evaluation criteria for resilience level of the OGI.

Numerical Range	Rating
[0, 0.3)	Low resilience
[0.3, 0.45)	Medium-low resilience
[0.45, 0.7)	Medium resilience
[0.7–0.85)	Medium-strength resilience
[0.85–1.0)	Strength resilience

The composite resilience index of China's OGI was calculated using Equations (1)–(7), with the results presented in Figure 2.

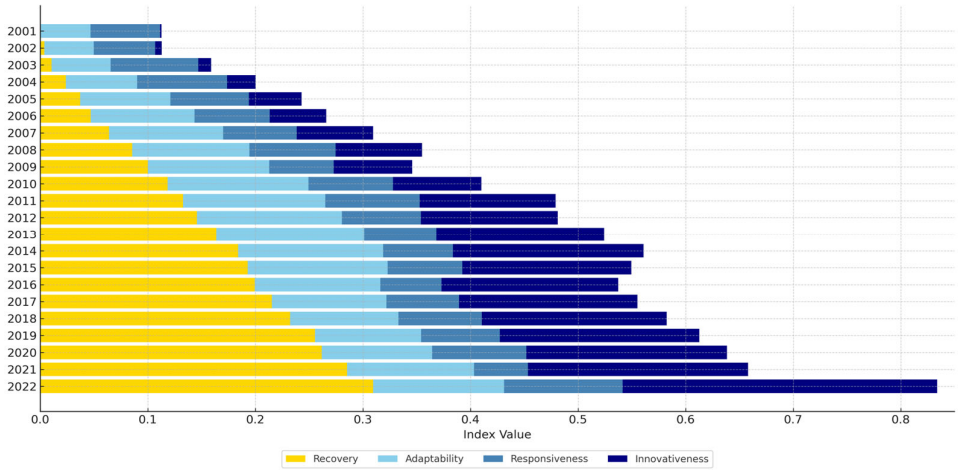

Figure 2. Evolution of resilience dimensions in China's OGI from 2001 to 2022.

From 2001 to 2022, the overall comprehensive resilience level of China's OGI showed an upward trend. Over the 22-year period, it increased from 0.1127 (2001) to 0.8338 (2022), with an average annual increase of 0.0328. This shows that the series of measures to enhance the resilience of China's OGI has achieved phased results, and the comprehensive resilience level has been greatly improved.

This article combines the OGI resilience rating evaluation standards and conducts analysis from two aspects: time series and the proportion of resilience in each dimension. In terms of time series analysis, according to the OGI resilience scores from 2001 to 2022, the development process is divided into three stages to reveal the evolutionary characteristics and driving factors of the resilience structure at different stages.

4.1.1. Analysis of Resilience Trend Evolution

(1) Phase of foundational accumulation. Between 2001 and 2007, the resilience score of China's OGI increased annually from 0.1127 to 0.3094, rising from low to lower-medium resilience. During this period, the industry began to build its foundation in recovery, adaptability, and responsiveness. Since 2005, supported by government policies and expanded resource development, investment in the sector increased significantly, leading to marked improvements in recovery and innovation resilience, and driving continuous growth in the overall resilience index. Although the outbreak of the global financial crisis in 2008 posed substantial challenges, the industry's resilience exhibited only minor fluctuations and then resumed an upward trajectory, rising from 0.3550 in 2008 to 0.5243 in 2013, with a notable milestone of 0.4791 in 2011. By that year, China's OGI had advanced into a medium resilience phase, demonstrating enhanced resilience and recovery capabilities.

(2) Stable Growth Phase. From 2014 to 2021, China's OGI experienced stable resilience growth; the composite resilience index increased from 0.5609 to 0.6580, representing an 18% rise. Substantial improvements were observed in both innovation resilience and recovery resilience. Facing volatility in global energy markets, oscillations in oil prices, and shifting economic conditions, the industry continuously promoted innovation by enhancing technological R&D and market responsiveness. From 2017 onwards, intensified investment in digital transformation and green-energy technologies steadily strengthened innovation resilience, resulting in the overall resilience index rising from 0.5609 in 2014 to 0.6127 by 2019. Beginning in 2020, the global pandemic delivered unprecedented market shocks, placing extraordinary pressure on the sector. In response, China's OGI adjusted production strategies, reinforced supply-chain management, and accelerated technological innovation, demonstrating robust adaptive capacity. Consequently, the composite resilience index

further increased from 0.6127 in 2020 to 0.6579 in 2021, sustaining its position within the moderate-resilience category.

(3) Mid-High Resilience Phase. In 2022, the composite resilience score of China's OGI rose to 0.8338—an increase of 26.71% compared to 2021—marking its entry into the mid-high resilience stage and demonstrating its competitiveness in the global energy market. The industry exhibited comprehensive improvements across recovery resilience, adaptability resilience, responsiveness resilience, and innovation resilience, reflecting its growing ability to adapt to shifts in global oil and gas markets and the challenges of energy transition. In the domains of green energy and low-carbon technology, the sector leveraged technological breakthroughs and policy support to address the global energy transition imperative, maintain high production efficiency, and accelerate its shift toward clean energy. China's OGI is therefore advancing toward global leadership, showcasing robust competitiveness and elevated resilience.

4.1.2. Analysis of the Proportion of Resilience in Each Dimension

The proportions of each resilience dimension, calculated from China's composite OGI resilience index, are presented in Figure 3.

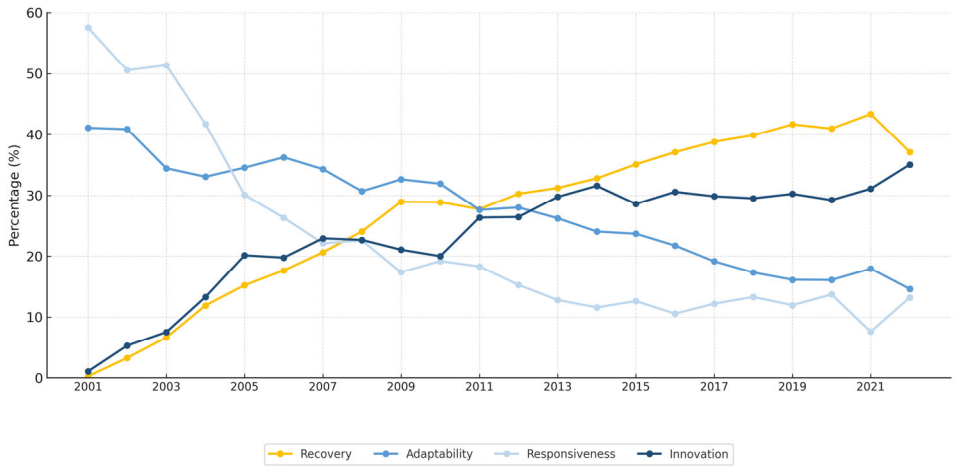

Figure 3. The proportion of each resilience dimension in China's OGI from 2001 to 2022.

From the changes in the proportion of resilience in each dimension from 2001 to 2022, it can be seen that the proportion of resilience in each dimension was significantly different between 2001 and 2008, and the order of resilience levels was relatively stable, as follows: responsiveness resilience > adaptability resilience > innovation resilience > recovery resilience. In the early stages, responsiveness resilience accounted for the highest proportion, about 57%, while recovery resilience was the lowest, at only 0.3%. This shows a huge difference between the two. Since 2008, the ranking of resilience has begun to change, and the proportions of responsiveness resilience and adaptability resilience have continued their previous downward trend, albeit with volatility. The importance of recovery resilience and innovation resilience in the overall picture continued to increase, and the differences in the proportions of the various components of comprehensive resilience narrowed. Subsequently, the proportions of resilience in each dimension remained in the range of 20% to 30%. Among them, recovery resilience surpassed adaptability resilience and ranked first in 2012, accounting for 30%. Its dominant position has become increasingly prominent. In 2022, the proportions of resilience in each dimension were: recovery resilience > innovation resilience > adaptability resilience > responsiveness resilience. Consolidating and strengthening the innovation resilience of the OGI will become the main trend in building the resilience of China's OGI in the future, while the downward trend in adaptability resilience also needs attention. The uncertainty of the external market and

the pressure of global energy transition have brought new challenges to the development of China's OGI.

4.2. Comprehensive Resilience Prediction

4.2.1. Simulation Scenario Setting

The study selects eight control variables from the OGI's resilience indicator framework, including full-time equivalent input of R&D personnel in the OGI, the OGI capital stock, and GDP. Four development scenarios are defined—baseline, slow population decline, rapid economic growth, and green transition—as presented in Table 3. By simulating and forecasting the industry's resilience levels from 2023 to 2032, the analysis compares changes across resilience dimensions under each scenario and proposes targeted policy recommendations.

Table 3. Parameter settings for different simulation scenarios.

Variable	Scenario 1	Scenario 2	Scenario 3	Scenario 4
Full-time Equivalent Input of R&D Personnel in the OGI		−5%		
OGI Capital Stock			+10%	
Gross Domestic Product			+10%	
Energy Industry Investment			+10%	
Carbon Emissions				−5%
CCUS Technology Adds New CO_2 Processing Capacity				+5%
Investment in Industrial Pollution Control Completed				+5%
Efficiency of Energy Conversion				+5%

Scenario 1: Natural state development scenario. In this scenario, the subsystems of the OGI maintain their current development status, and no intervention is made in the relevant variable parameters in the model. Based on historical statistical data and industry inertial development, this article obtains historical GDP, population, energy production, number of oil and gas employees, R&D investment, and other original data from the "China Statistical Yearbook (2001–2023)" and uses the "China Energy Development Annual Report (2022)" as the benchmark background for the annual operating status and trend forecasts in the oil and gas field. We use this information to run scenario simulations and to draw analogies with other development scenarios.

Scenario 2: Slow population decline scenario. In 2022, China experienced negative population growth for the first time, and the nation's population inflection point had been reached; by 2023, the total fertility rate had fallen to approximately 1.09—well below the replacement level of 2.1—and was among the lowest internationally. According to the China Population Development Forecast Report (2023), the working-age population (15–64 years) is projected to decline continuously from 2023 through 2035 [36]. The combined effects of population decline and population ageing are therefore expected to exert material downward pressure on labor supply and labor demand in the OGI, including the availability of R&D personnel. To capture this transmission pathway, we represent the demographic shock as a −5% adjustment to the full-time equivalent input of R&D personnel in the OGI. This −5% perturbation is grounded in neutral projections of working-age population decline reported by the National Bureau of Statistics and recent demographic forecasts, and it is specified as a moderate scenario shock—neither extreme nor trivial—intended to reflect a plausible mid-range impact of demographic change on R&D labor inputs.

Scenario 3: Rapid economic growth scenario. According to the "Outline of the 14th Five-Year Plan for National Economic and Social Development of the People's Republic of China and the Long-range Goals Through 2035", the annual average GDP growth target during the planning period is above 5%. This study defines a stress upper bound of +10% to represent a "high-growth" scenario. The choice of +10% as the upper bound is grounded

in the policy context of "seeking progress while maintaining stability" under the 14th Five-Year Plan and informed by historical experience. Within this high-growth scenario, GDP, OGI capital stock, and energy industry investment are each increased by 10%.

Scenario 4: Green transformation scenario. The 14th Five-Year Plan for the OGI proposes a green transformation path of "low-carbonization, cleanliness and high efficiency". To achieve this goal, the plan emphasizes accelerating the large-scale application of CCUS technology and promoting the integrated development of the OGI and new energy. The "China Carbon Peak and Carbon Neutrality Progress Report (2023)" proposes that carbon emission intensity in the industrial sector needs to decrease by 1.5–2% annually, providing a parameter basis for the "5% emission reduction" setting in this article. Based on this, a green transformation development scenario was set up, in which the CCUS technology added new CO_2 processing capacity [33], investment in industrial pollution control was completed, energy conversion efficiency was increased by 5%, and carbon emissions were reduced by 5%.

4.2.2. Model Verification

This paper uses MATLAB R2024b software and establishes a prediction model for the resilience of China's OGI from 2001 to 2022 based on the BP neural network model. In terms of data division, the training set accounts for 70% of the total data, and the test set accounts for 30%. Three historical data points are used as feature inputs. In order to improve the accuracy of the prediction, this paper adopts the method of randomly shuffling samples for training. The BP neural network regression fitting diagram is shown in Figure 4, and the simulated values, actual values, and error rates are shown in Figure 5 and Table 4. Judging from the simulated values, actual values, and error rates of the test set of China's OGI resilience from 2001 to 2022, except for the error rate of the resilience index of sample point No. 3 being higher than 10%, the absolute errors in other years are all below 10%. The R2 values of the test set and training set are both greater than 0.8, indicating that the simulated values of the model are close to the actual values, the model is reasonably constructed, and it has good behavior replication capabilities. The BP neural network model of China's OGI resilience constructed in this paper has good simulation performance and stability. It is a simulation model with high prediction strength. It can reflect the interactive relationship between the resilience level and the elements of the OGI system, and it is used to simulate and predict the development status of resilience in the OGI, providing an important basis for measuring changes in industry resilience.

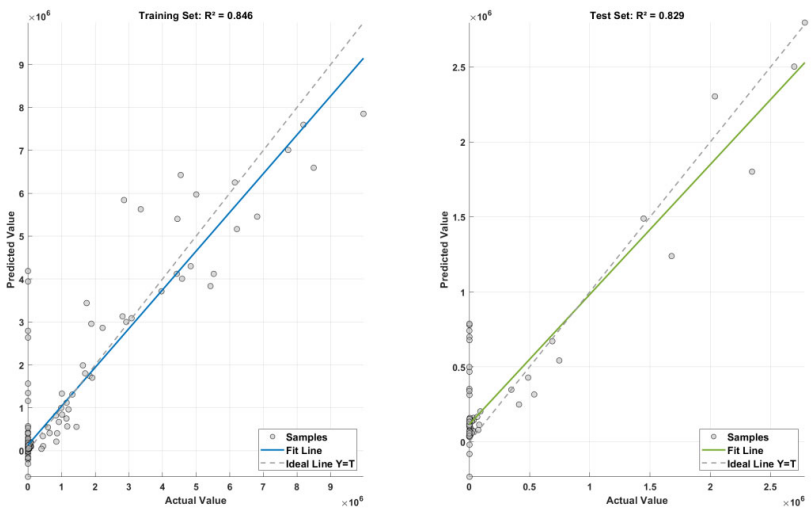

Figure 4. Neural network regression fitting diagram.

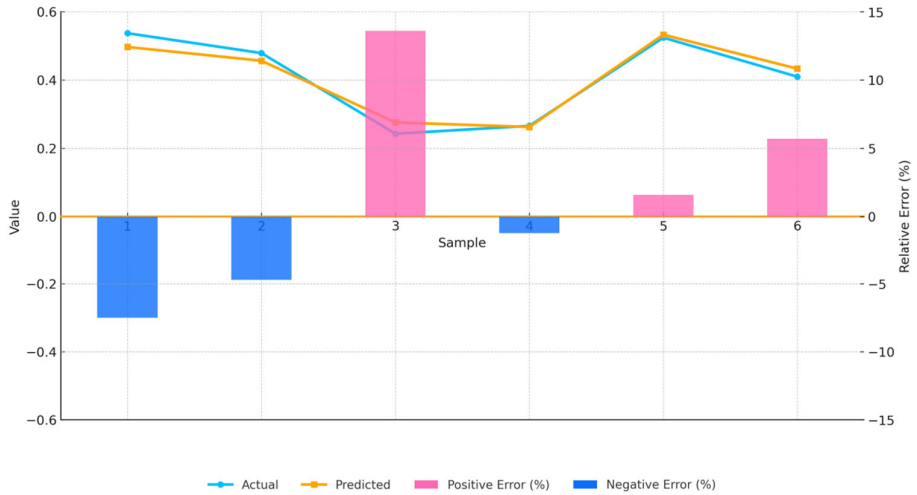

Figure 5. Comparison of predicted values and actual values of the test set.

Table 4. Test set test results.

Sample No.	Actual Value	Predicted Value	Error Rate
1	0.5373	0.4970	−7.5%
2	0.4791	0.4567	−4.7%
3	0.2430	0.2761	13.6%
4	0.2659	0.2627	−1.2%
5	0.5243	0.5329	1.6%
6	0.4100	0.4334	5.7%

4.2.3. Analysis of Prediction Results

The comprehensive resilience and the resilience scores of each dimension under four scenarios, namely, natural state development scenario, slow population decline scenario, rapid economic growth scenario, and green transformation scenario, were obtained by using the BP neural network model for scenario prediction, as shown in Table 5 and Figure 5.

Table 5. Comprehensive resilience of the OGI under multiple scenario simulations from 2022 to 2032.

Years	Natural State Development Scenario	Slow Population Decline Scenario	Rapid Economic Growth Scenario	Green Transformation Scenario
2023	0.790837	0.813497	0.796688	0.832620
2024	0.823568	0.810166	0.815364	0.895446
2025	0.862853	0.813556	0.846149	0.919345
2026	0.863821	0.798671	0.860490	0.896919
2027	0.879612	0.807450	0.869818	0.901176
2028	0.889159	0.813540	0.873668	0.911445
2029	0.875086	0.796124	0.872884	0.895265
2030	0.875334	0.794074	0.871738	0.897262
2031	0.867322	0.784874	0.870241	0.890631
2032	0.854439	0.771841	0.868693	0.879560

The results show that, taking 2022 as the benchmark, by 2032, the comprehensive resilience level of China's OGI will reach 0.8544, 0.7718, 0.8687, and 0.8796 under various development scenarios. Except for the slow population decline scenario, the comprehensive resilience of China's OGI will be further enhanced, entering the stage of strength resilience. The green transformation scenario can help China's OGI achieve strength resilience faster. Overall, except for the natural state development scenario, the resilience development

potential under the slow population decline scenario is the smallest, and the resilience development potential under the green transformation scenario is the largest. It can be seen that emphasizing the development goal of green transformation will be more helpful in resolving the contradiction between the development needs of China's OGI and uncertain interference, and it will have good development effects. Therefore, we should focus on the green transformation development path to maximize the comprehensive resilience of the OGI.

(1) Under natural conditions, the overall resilience of China's OGI showed a trend of slowly rising and then slightly falling, as shown in Figure 6. The score has been increasing steadily since 2023 and reached a peak of 0.8892 in 2028. It then fell slightly to 0.8544 in 2032, still within the strength-resilience range. Looking at it from different dimensions, the innovation resilience continues to increase, with its share rising from 37% in 2023 to 45% in 2032, gradually becoming the dominant force. This trend is highly related to China's efforts to strengthen oil and gas market-oriented reforms, improve resource allocation efficiency, and promote technological innovation in recent years. Adaptability resilience remains stable, while recovery and responsiveness show a downward trend due to market fluctuations and energy structure adjustments. In the short term, industry resilience is greatly affected by international oil price fluctuations and energy structure optimization; in the long term, it faces multiple uncertainties such as resource constraints, strengthened environmental governance, and carbon neutrality policies. Therefore, even if industry efficiency continues to improve, its stability and sustainability still need to be driven by policies and technology.

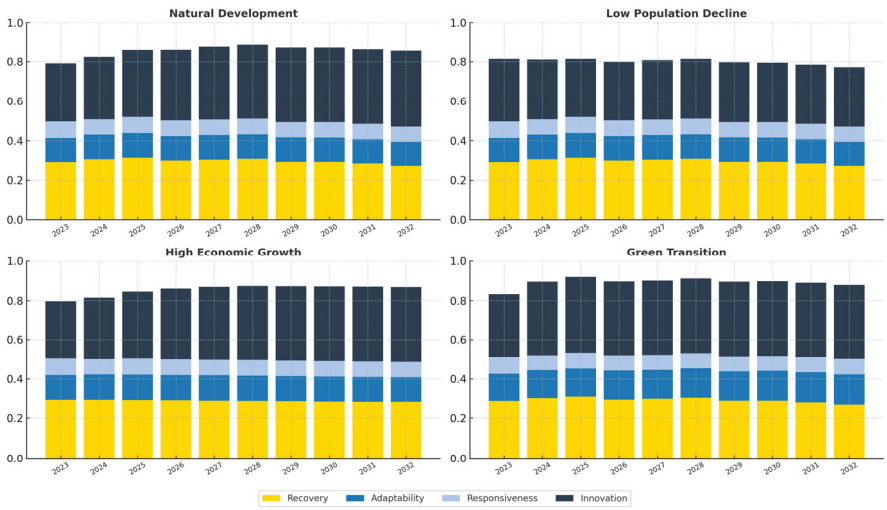

Figure 6. Prediction results of resilience of China's OGI by dimension from 2023 to 2032.

(2) Against the backdrop of a slow population decline, the resilience of the OGI continues to decline, from 0.8135 in 2028 to 0.7718 in 2032, remaining in the medium-strength resilience range. This scenario directly inhibits the improvement of innovation resilience by adjusting R&D personnel input to reflect the impact of population changes. As the labor supply and energy demand decrease simultaneously, companies' willingness to invest declines, market space is limited, and the momentum of technological progress and industrial transformation is suppressed. To deal with this scenario, the policy level can strengthen the response capabilities of enterprises through tax incentives, R&D subsidies, and digital upgrades. Oil and gas companies should accelerate intelligent transformation, reduce reliance on manual labor, and improve capital efficiency; from the perspective of energy structure, promoting the coordinated development of oil and gas and new energy

will help stabilize industry resilience and achieve structural balance in the context of population changes.

(3) Under the scenario of rapid economic growth, the resilience score increases from the baseline value of 2022 to 0.8605 in 2026 and further to 0.8687 in 2032—representing a total increase of 4.19%—while consistently remaining within the strength-resilience range. Economic expansion is strongly predictive of higher industrial recovery resilience, as indicated by its close association with increases in capital stock, energy investment, and total GDP. Industrialization and infrastructure construction are accelerating energy consumption and providing support for oil and gas. However, rapid growth is also accompanied by rising environmental pressure and increased carbon emissions, which restricts the sustainable development of the OGI. To resolve potential conflicts, we can guide the OGI to integrate with emerging fields such as advanced manufacturing and digital energy to increase added value. At the same time, we can build a sound carbon trading mechanism and green financial system, guide enterprises to transform into green enterprises, and optimize the oil and gas import structure to enhance the stability of industry resilience.

(4) Under the green transformation scenario, the resilience of the OGI will enter the strong resilience range first, and although there will be slight fluctuations after 2030, it will still remain at a relatively high level. This scenario focuses on improving adaptability and innovation resilience by adjusting key variables such as carbon emissions, CCUS processing capacity, and energy processing efficiency. Compared with the baseline scenario, it reflects the positive effects of low-carbon technology, energy structure adjustment, and environmental governance investment in enhancing industrial resilience. Although there were problems of rising costs and path dependence in the early stages of the transformation, with policy support, oil and gas companies have gradually achieved low-carbon and intelligent transformation. In this scenario, we should accelerate the research and development of CCUS and hydrogen energy, expand green financial support, and build an efficient new energy–oil and gas integration system to promote the resilience and competitiveness of the OGI in the global energy transformation.

5. Discussion and Limitations

This study recognizes several data-related limitations. First, owing to the incompleteness of historical records, certain indicator values were derived through interpolation, which may introduce potential measurement errors. Second, variations in statistical standards across different data sources could affect the overall consistency of the dataset. Third, long-term forecasting is inherently subject to uncertainty, as future economic, technological, and policy developments cannot be predicted with high precision. Consequently, the results should be regarded as indicative trends rather than exact quantitative outcomes. Nevertheless, the proposed framework offers a valuable reference for examining the resilience trajectory of the OGI. Future research could benefit from incorporating more comprehensive and high-frequency datasets, adopting advanced forecasting techniques such as machine learning-based hybrid models, and integrating policy scenarios to enhance the robustness and practical relevance of resilience assessments.

6. Conclusions

Based on the relevant data of the OGI from 2001 to 2022, this paper takes into account the current status of industry recovery resilience, adaptability, responsiveness, and innovation, constructs a resilience evaluation index measurement system for China's OGI, and conducts a comprehensive assessment of the changes in the resilience of the OGI. The BP neural network is used to calculate the data indicators from 2001 to 2022, and multi-

scenario predictions are made for China's OGI under different scenarios, such as natural development, slow population decline, rapid economic growth, and green transformation and development. The conclusions are as follows:

(1) Recovery and innovation resilience have gradually become dominant. From 2001 to 2022, the overall recovery, adaptability, responsiveness, innovation, and comprehensive resilience levels of China's OGI showed a synchronous upward trend, gradually improving from low resilience to the medium-strength resilience stage. Among them, the proportion of recovery and innovation resilience showed a strong growth momentum, while the proportion of adaptability resilience and responsiveness showed a clear downward trend. China's OGI has not yet entered the stage of strength resilience.

(2) The proportion of each dimension of resilience is still in a stage of fluctuation and continuous adjustment. Two development trends of dimensional resilience can be observed. The first one, represented by adaptability resilience and responsiveness resilience, decreases year by year, while the other one, represented by recovery resilience and innovation resilience, increases year by year. There are differences in the increase in resilience in different dimensions, which has gradually shifted from being led by responsiveness resilience to being led by innovation resilience, and innovation resilience has gradually become the development focus of the future resilience improvement process. The proportion of resilience in each dimension shows a steady upward trend. Meanwhile, the resilience and coordination within the OGI have been enhanced, and China's OGI is transitioning from a structurally stable system to an innovation-driven one.

(3) The green transformation and rapid economic growth scenarios show strong potential for resilience improvement. Through four scenario simulations, it was found that compared with 2022, by 2032, the comprehensive resilience level of China's OGI will increase by 2.48%, 4.19%, and 5.49%, respectively, under the natural development, rapid economic growth, and green transformation scenarios; while under the scenario of slow population decline, the resilience level will decline. The research results show that optimizing energy structure, promoting technological innovation, and building a diversified development path can help significantly improve the resilience of the OGI in a complex risk environment. Green and low-carbon transformation is not only an important support for achieving the national energy strategic goals but also a key direction for enhancing industrial resilience and international competitiveness.

(4) Policy and industry recommendations. In light of the finding that the green-transition scenario yields the greatest resilience gain (5.49% by 2032), China's OGI should strengthen technological R&D and market responsiveness to drive continuous innovation. Concrete measures include increasing investment in digital transformation and green energy technologies, accelerating the industrial application and demonstration of intelligent technologies, and improving institutional mechanisms for supply chain optimization and upgrading. At the policy level, sustained support for green and low-carbon development is needed through fiscal and tax incentives, refinement of unconventional oil and gas taxation, and the introduction of targeted subsidies. Furthermore, enhancing top-level design and establishing a comprehensive standard system aligned with national strategies can guide the industry toward environmentally sustainable development while also promoting resilience and security [37]. Collectively, these measures will reinforce recovery, adaptability, responsiveness, and innovation, ensuring that the oil and gas industry evolves into a more resilient, competitive, and sustainable sector.

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Abbreviations

The following abbreviations are used in this manuscript:

OGI	Oil and Gas Industry
BP	Backpropagation
CRA	Critical Resilience Assessment
ML	Machine Learning
AI	Artificial Intelligence
EGARCH	Exponential Generalized Autoregressive Conditional Heteroskedasticity
CCUS	Carbon Capture, Utilization, and Storage
GDP	Gross Domestic Product
R&D	Research and Development
Min{}	Minimum function
Max{}	Maximum function
lnx	Logarithmic function

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